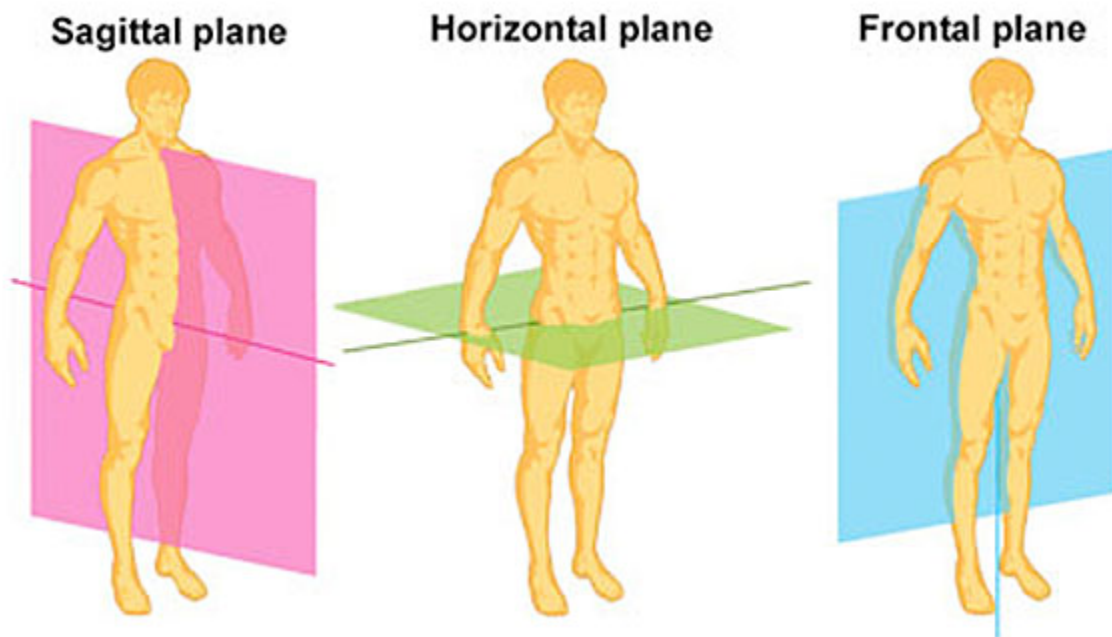


Shoulder Flexion

Performance:

- Patient stands upright against or near a wall
- Feet shoulder width apart, back straight
- One arm resting at side, elbow fully straight
- Palm facing inward toward the body (neutral grip)
- Patient raises the arm forward and upward in a smooth arc
- Continues to comfortable end range (90° horizontal or higher)
- Holds briefly at the top
- Lowers back down slowly and with control
- Returns arm fully to side before next rep

The movement is **entirely in the sagittal plane** (the body is divided into two equal halves **vertically**) — the arm travels forward and upward directly in front of the body, not out to the side. This is the key distinction between shoulder flexion and shoulder abduction. Both look like raising an arm but in completely different planes.

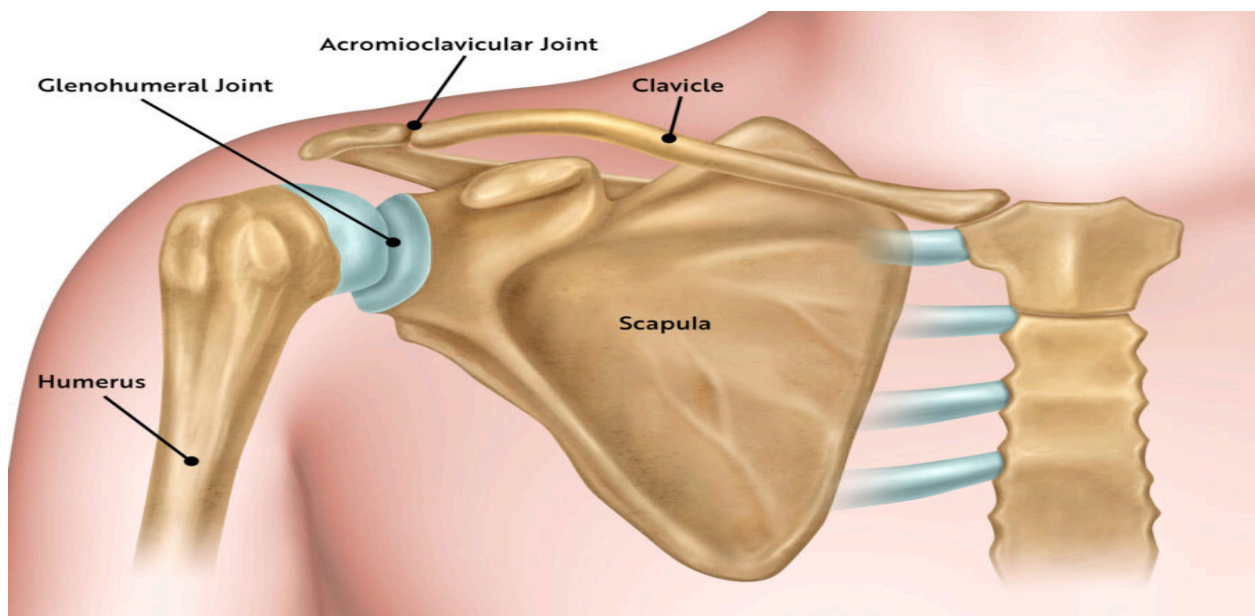


Camera:

- View - strict side view, perpendicular to patient
- Camera height - shoulder height of patient (~130-150cm from floor)
- Camera distance - 2 to 2.5 metres from patient
- Background - plain wall, minimal clutter

- Lighting - even, diffuse, light source in front of patient. Not behind (no window behind patient)
- Clothing - fitted short-sleeved or sleeveless top
loose clothing around the shoulder hides the landmark and causes MediaPipe drift
- Patient position - side-on to camera, moving arm closest to camera. The other arm relaxed at the side.

For shoulder flexion the **primary landmarks** are **shoulder, elbow and wrist**. When the arm is at rest these landmarks are between waist and shoulder height, and when fully raised they go above the head. A camera at **shoulder height captures the full arc symmetrically** — the arm starts roughly level with the camera and travels upward through and above the frame's midpoint. A camera at hip height looking upward introduces perspective distortion — the arm appears to shorten as it rises because you are increasingly looking at it from below at an angle.



Factors that might affect the starting position:

- 1) Thoracic Kyphosis (The shoulder landmark is directly affected. As a result, the shoulder flexion angle which is calculated as hip-shoulder-elbow will be different for each person)
 - The natural forward rounding of the upper spine. Every person has some degree of thoracic kyphosis, it is normal. But the degree varies enormously:
 - Young healthy adult: mild kyphosis, relatively upright
 - Elderly patient: often significant kyphosis
 - Post-surgical patient: may have altered posture from guarding

- Office worker: often moderate kyphosis from desk posture
- Shoulder flexion angle is calculated as: **hip - shoulder - elbow**
- The shoulder landmark position in the frame is directly affected by how upright or rounded the upper back is. A patient with **significant kyphosis** will have their **shoulder sitting forward and downward relative to their hip** compared to an upright patient — even when both patients are standing in their natural resting posture.

1) Upright patient

- Hip at (400, 500)
- Shoulder at (390, 300) ← shoulder nearly directly above hip
- Shoulder_angle at rest $\approx 175^\circ$

2) Kyphotic patient

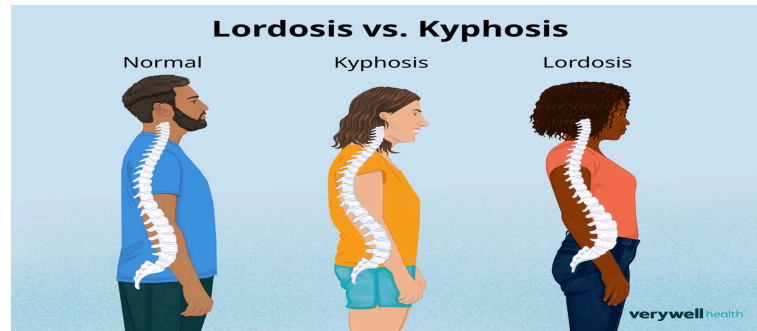
- Hip at (400, 500)
- Shoulder at (350, 320) ← shoulder sits forward of hip
- Shoulder_angle at rest $\approx 160^\circ$

So the S1 baseline shoulder angle — the resting arm-at-side starting position — will be systematically different between these two patients even before any movement happens. This is not an error. It is just anatomy. But your model will see different S1 baselines and needs to not misinterpret this as incorrect form.

How to handle it:

The same principle as ROM in heel slides - express your features relative to each patient's own S1 baseline rather than as absolute angles. A patient whose arm starts at 160° (due to kyphosis) and raises to 80° has achieved 80° of flexion. A patient who starts at 175° and raises to 95° has also achieved 80° . The absolute angles differ but the movement is equivalent. (The key here is that instead of the starting/ resting position - the focus needs to be on the movement.)

For Kyphosis or Lordosis the most important feature here would be the trunk lean in the resting position or S1 (this will tell us the extent of the condition and thereby help us to calculate the baseline). The ROM will be calculated based on this initial baseline. (Initial baseline/ trunk lean will be different for a normal patient compared to a patient with Kyphosis or Lordosis.)



2) Scapular Position & Mobility

The scapula (shoulder blade) sits on the back of the ribcage and its resting position varies between individuals. This is the closest equivalent to the Q angle for the shoulder.

Normal scapular position - flat against ribcage, centred

Winged scapula - medial border lifts away from ribcage

Protracted scapula - shoulder blade pulled forward and around the ribcage (common in rounded shoulders)

Elevated scapula - shoulder sitting higher on one side

Why it affects your angles:

MediaPipe's shoulder landmark (number 11 or 12) localises to the **visible surface** of the shoulder from the side — roughly the deltoid region. The actual glenohumeral joint (where the arm bone meets the shoulder blade) sits deeper inside the body. For patients with forward-protracted shoulders, the visible surface landmark is positioned further forward relative to the true joint centre. This creates a systematic offset in every angle that uses the shoulder landmark.

3) Arm Length Variation

People have different arm lengths — specifically different ratios of upper arm (humerus) to forearm (radius and ulna) length.

Your elbow angle and wrist position features depend on where the elbow and wrist landmarks are in the frame. A patient with a proportionally longer forearm will have their wrist landmark appearing further from their elbow in the frame. Since your angle calculations are projection-based, longer limbs produce slightly different raw angle readings for the same actual joint angle at the shoulder.

How to handle it:

This is actually largely handled by your normalisation approach already. Because your pelvic gap normalisation introduced the principle of dividing by torso length, and because your angle calculations inherently normalise by vector lengths, most of the absolute limb-length variation

cancels out. The main residual effect is at near-collinear positions — exactly as with the knee — where small absolute differences in landmark placement produce proportionally large angular errors.

4) Glenohumeral Joint Hypermobility

Some patients have naturally hypermobile shoulder joints — they can raise their arm significantly above 180° (past vertical, continuing behind the head) without any pathology. This is particularly common in younger female patients and patients with generalised hypermobility syndrome.

Why it matters:

Your rep detection looks for peaks at the resting arm-at-side position (~170-180°). A hypermobile patient who raises their arm to 20-30° past vertical will have a very large rep range — the shoulder angle going from 175° down to perhaps 10-20°. This is a range of 155-165°, significantly larger than what you might see in a typical post-surgical patient achieving 90-100° of flexion.

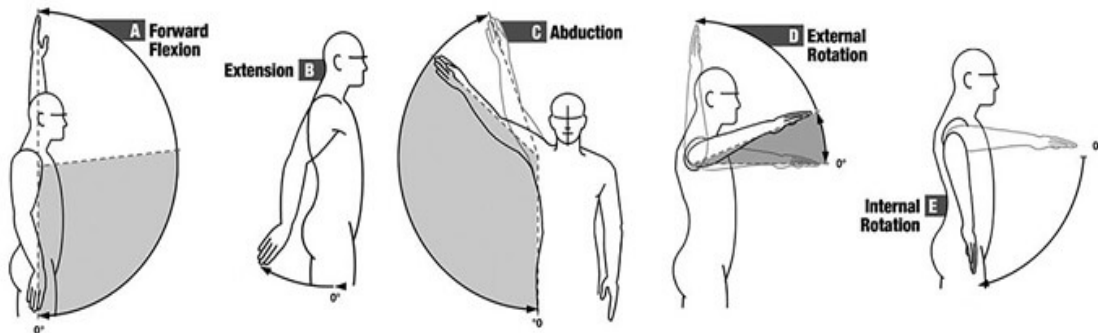
Variable	Knee equivalent	Impact on your angles
Thoracic kyphosis	Q angle	S1 baseline shifts per patient
Scapular protraction	Q angle	Shoulder landmark offset
Arm length ratio	Leg segment ratio	Minor, largely self-correcting
Glenohumeral hypermobility	Genu recurvatum	Large rep range variation
Bilateral asymmetry	Valgus/varus	Affected arm behaves differently

Key Angles:

1) Shoulder Flexion Angle (Primary Tracking Angle)

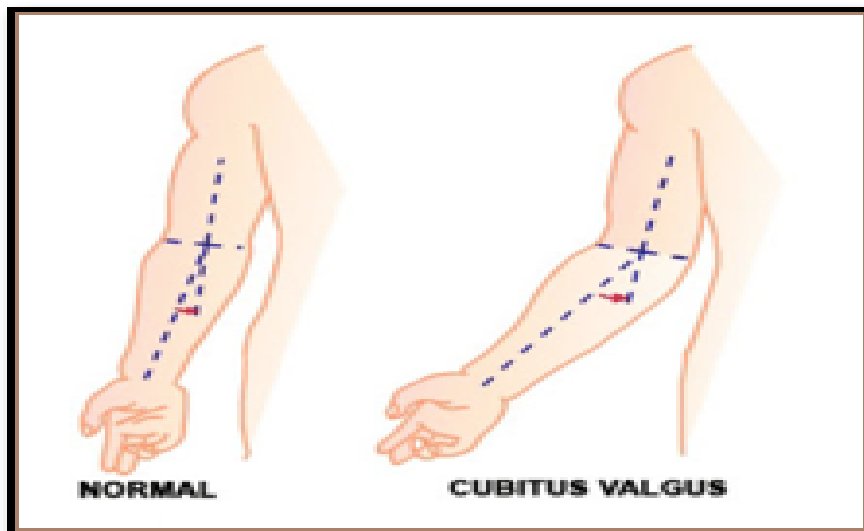
- This is the tracking angle for rep detection. The peaks correspond to the arm at rest by the side (180 degree. Therefore, similar to the heel slides - the extended positions or peaks form the rep boundaries and the valleys in the data form the midpoint of the rep.)
- According to the [American Academy of Orthopaedic Surgeons \(AAOS\)](#), a normal, healthy shoulder flexion angle is **180 degrees**, meaning the arm can point straight up toward the ceiling.
- **160° – 170°**: Generally considered functional and healthy for most adults in everyday activities.

- **Less than 150°:** Often indicates joint restriction, muscle tightness (such as the latissimus dorsi), or a stiff thoracic spine



2) Elbow Angle (primary correctness feature)

- Landmarks → shoulder (12/11) - elbow (14/13) - wrist (16/15)
- Should stay → ~175-180° throughout the entire movement
- This is your most important universal correctness feature
- A bending elbow is always wrong regardless of how high the arm goes



3) Trunk Lean Angle (Compensation)

Landmarks → vertical reference - hip - shoulder

Should stay → near 0° (upright) throughout the movement

Compensation pattern → patient leans trunk backward as arm rises

this is the primary error in shoulder flexion

it uses trunk extension to artificially increase

apparent shoulder ROM without true shoulder mobility

4) Wrist position relative to elbow — secondary

Landmarks → elbow (14/13) - wrist (16/15)

Should stay → wrist in line with or above elbow throughout raise
wrist dropping below elbow = compensation pattern
suggests elbow bending or wrist flexion for momentum

5) Shoulder elevation (shrugging) — secondary

Track the vertical position of the shoulder landmark (12/11)

As the arm raises, the shoulder itself should not rise significantly
toward the ear — this is called scapular elevation
or shrugging, and is a compensatory pattern that
recruits the trapezius instead of the true shoulder
flexors

Detected by → shoulder_y position compared to S1 baseline
rising shoulder_y during S2 = shrugging compensation

States:

S1 → arm at side, shoulder ~170-180° (starting position)

S2 → arm rising, shoulder angle decreasing (ascending phase)

S3 → peak height, shoulder angle at minimum (held at top)

S4 → arm lowering, shoulder angle increasing (descending phase)

Common Errors

1) Error 1 — Trunk lean backward (most common)

- What happens - patient arches lower back and leans trunk backward to swing the arm upward using momentum rather than true shoulder flexor strength.
- Why it happens - weak anterior deltoid or rotator cuff fatigue
- Feature - trunk_angle deviates significantly in S2 and S3 from the S1 baseline
- Universal? - yes, always wrong regardless of height achieved

2) Error 2 – Elbow Bending

- What happens - elbow flexes partway through the raise
- Why it happens - patient shortens the lever arm to make the lift easier. A shorter arm is mechanically easier to raise than a straight arm.
- Feature - elbow_std high in S2 and S3. elbow_min drops noticeably below 175°
- Universal? - yes, always wrong

3) Error 3 – Shoulder Shrugging

- What happens - patient elevates the shoulder toward the ear as the arm rises, recruiting the upper trapezius.
- Why it happens - rotator cuff weakness causing the deltoid to pull the whole shoulder complex upward rather than purely rotating the arm at the glenohumeral joint.
- Feature - shoulder_y position rises during S2 relative to its S1 baseline (remember y decreases as you go up in the frame, so shoulder_y decreasing means it is rising toward the ear)
- Universal? - yes, always wrong

4) Error 5 – Uncontrolled Lowering

- What happens - patient drops the arm quickly on the way down rather than lowering with eccentric control.
- Why it happens - fatigue, or the patient not understanding the eccentric component is therapeutically important.
- Feature - shoulder_std high in S4. S4_duration is significantly shorter than S2_duration. speed asymmetry between ascent and descent
- Universal? - yes, always wrong

Relative to S1 is needed when:

The starting position varies meaningfully between patients
due to anatomy, posture or condition severity
AND the absolute value of the feature would therefore mean
different things for different patients

Relative to S1 is NOT needed when:

The starting position is physically constrained to be
approximately the same for everyone (leg flat on bed)
OR the feature is already measuring change/quality/ratio
and has no dependency on absolute starting position

For supine exercises — heel slides, straight leg raise, bridging — the starting position is largely physically constrained by gravity and the bed surface. Relative features add less value here than for standing exercises where posture freely varies.

For standing exercises — shoulder flexion, wall slides, squats — postural variation is real and meaningful. Relative features genuinely help here.